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SMOC : A MODULAR, SELF-CALIBRATING SCARA ROBOT DESIGN

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MECHANICAL DESIGN & CAD ARCHITECTURE

Proof of Concept — 5-DOF Desktop SCARA Robotic Manipulator

Project Resources and Explainer Video: https://shorturl.at/lppgW

1. INTRODUCTION

This document presents the mechanical design and CAD architecture developed for the SCARA (Selective Compliance Assembly Robot Arm) robotic manipulator, forming the structural and kinematic foundation of the larger Anveshan 2026 SCARA Robot project. While the overall submission integrates simulation (Unity3D/ROS2), control architecture (MATLAB/Simulink closed-loop control), and embedded PCB design as parallel sub-modules, this section focuses exclusively on the physical robot: its degrees of freedom, joint architecture, structural design, transmission systems, and material strategy, as realised in SolidWorks CAD.

The objective of this design is to produce a compact, desktop-scale, precision manipulator capable of pick-and-place, small-part assembly, and vision-guided manipulation tasks, while remaining manufacturable through a combination of additive manufacturing and off-the-shelf motion components. The design deliberately favours a modular, serviceable architecture so that the same CAD platform can transition from a 3D-printed proof-of-concept to a metal, field-deployable unit without a structural redesign.

FULL ASSEMBLY — ISOMETRIC RENDER

Fig. 1 — Full assembly isometric render of the 5-DOF SCARA manipulator.

2. ROBOT ARCHITECTURE & KINEMATIC CONFIGURATION

The manipulator is a custom 5-degree-of-freedom desktop SCARA, following a Prismatic–Revolute–Revolute–Revolute–Prismatic (P-R-R-R-P) joint chain. This configuration provides the classic SCARA workspace — high stiffness along the vertical (Z) axis and selective compliance in the horizontal (X-Y) plane — while an added wrist revolute joint and a rack-and-pinion gripper give the arm full end-effector orientation and precise grasping control. The chain proceeds from the base, through the vertical lead-screw column, the shoulder and elbow revolute joints, the wrist rotation, and terminates at the parallel gripper.

Joint	Type	Actuation	Primary Function
DOF 1 — Z Axis	Prismatic	Lead-screw linear actuator + linear guide rail	Vertical approach, pickup, placement, height adjust
DOF 2 — Shoulder	Revolute	BLDC motor, vertical axis	Primary arm sweep, workspace coverage
DOF 3 — Elbow	Revolute	BLDC motor, vertical axis	Reach extension, fine positioning
DOF 4 — Wrist	Revolute	BLDC motor, vertical axis	End-effector orientation, placement alignment
DOF 5 — Gripper	Prismatic	Rack-and-pinion, parallel jaws	Grasp, release, precision manipulation

Table 1. Joint-wise breakdown of the 5-DOF SCARA kinematic chain.

3. STRUCTURAL DESIGN

Base & Electronics Enclosure

The base is designed as an enclosed housing with a removable cover, internally compartmentalised to house the PCB, motor drivers, wiring harnesses, and power electronics. Enclosing all electronics within the base lowers the overall centre of gravity, improves wiring cleanliness, and simplifies maintenance access without disturbing the arm's kinematic assembly above it.

Vertical Column (Z-Axis)

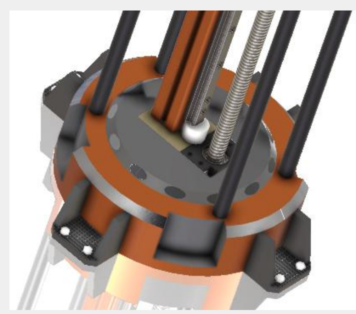
The column integrates the linear guide rail, lead screw, drive motor enclosure, and a compact sliding carriage into a single rigid structural member. This carriage carries the shoulder joint and the remainder of the arm, so the column must simultaneously support axial (vertical) loads and resist the bending moment introduced by the fully extended arm — driving the choice of a ribbed, large-cross-section column profile.

Links (Upper & Fore-arm)

Both arm links use a hollow, ribbed shell geometry with rounded edges and a deliberately large bending depth. This is a weight-optimisation strategy: removing material from the neutral bending axis while retaining depth preserves stiffness-to-weight ratio, directly reducing the rotational inertia the shoulder and elbow BLDC motors must overcome during acceleration.

Wrist & End Effector

The wrist is a compact rotational stage that orients the gripper prior to grasp. The end effector is a parallel-jaw gripper actuated through a rack-and-pinion mechanism, chosen for symmetric jaw travel, positive mechanical engagement, and repeatable grip force — suited to PCB handling, small mechanical parts, and laboratory-sample manipulation.



BASE & Z-COLUMN CLOSE-UP

Fig. 2 — Base enclosure and Z-axis lead-screw column detail.

SHOULDER / ELBOW ACTUATOR CLOSE-UP

Fig. 3 — Revolute joint actuator (BLDC motor housing) detail.

4. MECHANICAL TRANSMISSION SUMMARY

- Linear axis (Z): Lead screw + linear guide rail + carriage — high positioning accuracy, low backlash, strong rigidity for vertical loads.
- Rotary axes (Shoulder / Elbow / Wrist): Direct BLDC-driven revolute joints in a coaxial arrangement with hidden motor housings — compact packaging and minimal exposed components.
- Gripper: Rack-and-pinion transmission — symmetric jaw motion, positive engagement, simple and repeatable.

5. ROBOT DIMENSIONS & WORKSPACE ENVELOPE

Values below are placeholders — insert final measured/CAD-derived figures.

Overall footprint (Base ⌀)	300 ± 5 mm
Total height (Z fully retracted)	100 ± 5 mm
Link 1 length (Shoulder–Elbow)	200 ± 5 mm
Link 2 length (Mid Extension)	175 ± 5 mm
Link 3 length (Mid Extension–Wrist)	150 ± 5 mm
Maximum horizontal reach	505 ± 5 mm
Maximum Payload	1.25 ± 0.05 kgs (Estimated for CAD (PLA) Build) 11.50 ± 0.25 kgs (Estimated for CCN (Aluminium) Build)

6. WORKSPACE & MOTION CHARACTERISTICS

The manipulator generates a cylindrical working volume: vertical reach is defined by the Z-axis stroke, and horizontal reach is defined by the two-link SCARA arm's combined length. The design exhibits high stiffness along Z with selective compliance in the X-Y plane — the defining SCARA trait that makes the arm naturally forgiving for vertical insertion tasks (e.g., PCB/connector assembly) while remaining precisely positioned laterally. The kinematic layout targets fast point-to-point positioning, smooth trajectory execution, high repeatability, and continuous wrist rotation, in line with the trajectory-planning and control-loop requirements defined by the control architecture sub-module.

7. DESIGN LANGUAGE & MANUFACTURABILITY

Visually, the platform follows a hidden-actuator, minimal-fastener design language: motor housings are enclosed, fillets are consistent across all structural parts, and the arm maintains symmetric proportions with an orange-and-black colour scheme intended to read as a finished product rather than a prototype rig.

From a manufacturability standpoint, the CAD is built as a modular assembly — a serviceable base enclosure, independently swappable link components, fully enclosed electronics, and provision for internal cable routing through the column and links. This structure is suited to additive manufacturing (FDM/SLA) of the structural shells, combined with off-the-shelf linear guides, lead screws, bearings, and fasteners for the motion-critical interfaces.

8. MATERIAL STRATEGY: PROTOTYPE VS. DEPLOYMENT

Material selection follows a two-phase strategy aligned with the project's development timeline: an early, low-cost, 3D-printed prototype for kinematic and control validation, followed by a metal/composite deployment-grade build once the design is proven through the testing rounds.

Phase	Structural Components	Motion / Hardware Components
Phase 1 — Prototype / Testing	PLA (FDM 3D-printed) — base shell, links, wrist housing, gripper jaws	Steel lead screw & linear rail, standard bearings, off-the-shelf BLDC motors and fasteners
Phase 2 — Final Deployment	Aluminium (6061-T6) for links & column — machined/CNC; Steel for high-load base and joint interfaces	Precision ground steel lead screw, hardened linear rails, rubber gripper jaw inserts for compliant grasp, sealed bearings

Table 2. Material progression from prototype to field-deployable configuration.

PLA is selected for the initial prototype phase due to its dimensional stability, ease of FDM printing, low cost, and adequate stiffness for validating kinematics, motion range, and control-loop tracking without committing to expensive tooling. For final deployment, structural links and the column transition to machined aluminium (6061-T6) for its high strength-to-weight ratio and corrosion resistance, while high-load interfaces — the base plate, joint shafts, and fasteners — move to steel for durability under repeated cyclic loading. Rubber inserts are introduced at the gripper jaw contact faces to increase grip friction and provide compliant, damage-safe handling of delicate PCB and electronic payloads.

9. MECHANICAL CHARACTERISTICS

- Payload category: Desktop precision manipulator — PCBs, sensors, electronic modules, small mechanical parts, laboratory samples.
- Compliance: High Z-axis stiffness with selective X-Y compliance, suited to assembly-style insertion tasks.
- Repeatability & motion: Optimised for fast positioning, smooth trajectories, and continuous wrist rotation, in line with the quintic trajectory planning used in the control sub-module.
- Primary material (prototype): 3D-printed PLA polymer shells with metal shafts, bearings, and fasteners.
- Primary material (deployment): Aluminium/steel structural members with rubber-faced gripper jaws.

10. TARGET APPLICATIONS

- PCB pick-and-place and electronic component handling
- Laboratory sample transfer and pick-and-place demonstrations
- Vision-guided manipulation with camera integration
- Desktop automation, educational robotics, and AI/ML manipulation research
- ROS 2 / MoveIt integration and human-robot interaction demonstrations

GRIPPER / END-EFFECTOR CLOSE-UP

Fig.4 — Parallel-jaw gripper with rack-and-pinion actuation.

WRIST ASSEMBLY CLOSE-UP

Fig.5 — Wrist rotation stage detail.

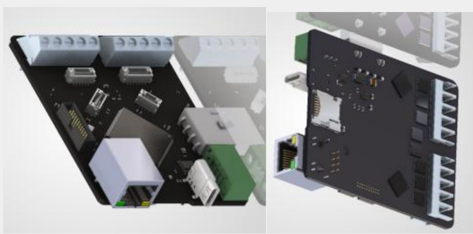
ELECTRONICS STACK & CONTROL ARCHITECTURE

Proof of Concept — Distributed Embedded Control & Self-Calibrating Sensor Fusion for the SCARA Robotic Manipulator

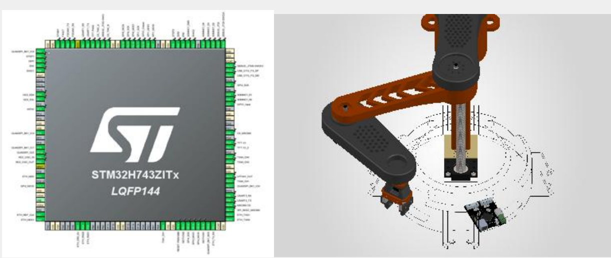
Sub-module: Embedded Electronics, Actuator Network & Real-Time Control Loop

1. INTRODUCTION

This document presents the embedded electronics stack and real-time control architecture for the SCARA manipulator — the sensing, actuation, and closed-loop intelligence layer of the Anveshan 2026 SCARA Robot project, built as a distributed hierarchical system that separates high-level supervisory decision-making on Vesper from deterministic, low-level actuator control on each joint node.



FULL ASSEMBLY — ELECTRONICS INTEGRATION SHOT (base cutaway / full robot render showing PCB & sensor placement)



STM32H743 Pin Configuration and PCB Wireframe in CAD

Fig. 1 — Full assembly render showing electronics stack integration within the base enclosure and along the arm.

2. SYSTEM CONTROL ARCHITECTURE

At the centre of the system is the Vesper Main Controller, built around an STM32H743ZIT6, which serves as the supervisory controller. Rather than directly driving motors, the STM32H7 executes computationally intensive tasks — sensor fusion, recursive state estimation, inverse kinematics, trajectory generation, and high-level closed-loop control — and manages external communication over Ethernet, USB Type-C, SPI, UART, I²C, and CAN-FD.

Motion actuation is performed by three independent Actuator Nodes, each built around an STM32G431. Every node integrates a TMC9660 Field-Oriented Control (FOC) driver, an MSC500 absolute magnetic encoder, an IMU, and a CAN-FD interface — the encoder is coupled directly to the TMC9660 for minimal-latency current, velocity, and position control, while encoder position, actuator status, and IMU data are streamed to Vesper over CAN-FD.

3. VESPER MAIN CONTROLLER

Vesper ONE is the single supervisory board of the system — every non-actuator sensor in the design terminates here, and no other board carries a direct external sensor connection. It is built around an STM32H743ZIT6, whose Arm Cortex-M7 core runs at up to 480 MHz with a double-precision FPU and hardware Chrom-ART acceleration, backed by up to 2 MB of flash and 1 MB of SRAM.

Alongside the H743, Vesper integrates a Trinamic TMC5160 stepper driver to control the Z-axis lead-screw motor. Its StealthChop2 voltage-mode chopper ensures quiet, low-vibration operation, while 256-step MicroPlyer interpolation provides smooth motion from coarse step inputs. The integrated stallGuard4 sensorless load detection enables automatic end-of-travel detection, allowing Z-axis recalibration without requiring physical limit switches.

Vesper communicates via an ADIN1200 Ethernet PHY for vision data and a MAX33012EASA+ CAN-FD transceiver for actuator, encoder, and IMU communication. High-bandwidth sensors, including the PMW3389 and MSC500, connect directly to Vesper, centralizing sensor fusion while keeping actuator nodes lightweight.

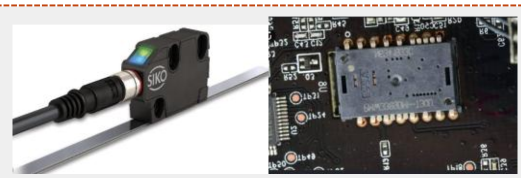
4. SENSOR FUSION FOR STATE ESTIMATION & SELF-CALIBRATION

The core contribution of this sub-module is a continuous self-calibration layer that keeps the robot's internal state accurate under drift, wheel slip, and mechanical offset — without requiring manual recalibration between runs. Four independent sensing sources feed a recursive state estimator running on the Vesper STM32H743:

- **Joint & actuator encoders (all three nodes)** — each TMC9660 is direct-coupled to its local encoder for closed-loop current, velocity, and position control; raw counts are also streamed to Vesper over CAN-FD as ground-truth joint feedback, independent of what was commanded.
- **PMW3389 optical motion sensor — base offset** — works like a high-speed optical mouse sensor: it images the surface beneath the base at a high frame rate and cross-correlates successive frames to compute incremental X-Y displacement. Integrating this over time gives the accumulated planar offset and rotation of the base caused by wheel slip or a mechanical bump.
- **MSC500 magnetic encoder — height measurement** — an absolute magnetic (Hall/AMR-based) linear encoder reading a magnetised scale along the Z-axis lead-screw column. It gives a non-contact, wear-free readout of the carriage's absolute height above the base at all times, rather than an incremental count that can drift.
- **IMU (per actuator node)** — per-node linear acceleration and angular velocity, providing joint orientation, base attitude, and dynamic motion state.
- **Vision system (external)** — object coordinates from an external vision system, transmitted via the ADIN1200 Ethernet PHY or USB Type-C.

Vesper compares the expected carriage height (calculated from the Z-axis lead-screw pitch and stepper turns) against the MSC500's measured height. Any discrepancy exceeding a set threshold is immediately fed into the actuator's FOC loop to compensate for missed steps, backlash, or mechanical drift before the next motion command.

Concurrently, an STM32H743 runs an EKF/EWMA filter to fuse all five data streams into a single, continuously corrected state estimate. By actively compensating for noise and drift, this highly accurate profile (covering position, orientation, and dynamics) feeds directly into inverse kinematics and trajectory planning, ensuring mechanical errors never propagate downstream.



MSC500 and PMW3389 Sensors

CONTROL, POWER & SOFTWARE ARCHITECTURE

Proof of Concept — Closed-Loop Control, Power Electronics, Motor Drive & Perception Stack for the SCARA Robotic Manipulator

Sub-module: MATLAB/Simulink Control Loop, Power Flow, FluxCruiser Motor Driver & Unity3D/ROS2 Vision Stack

1. CONTROL ARCHITECTURE

The developed MATLAB/Simulink model implements a complete closed-loop control architecture for a robotic manipulator to achieve accurate and smooth trajectory tracking. The system begins by accepting predefined Cartesian coordinates along with their corresponding timestamps, which define the desired path of the end effector. These Cartesian waypoints are converted into joint-space waypoints using an analytical Inverse Kinematics algorithm, enabling the manipulator to execute the desired motion.

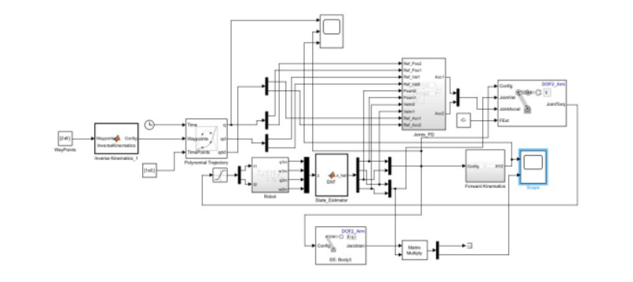


Fig 1.1 — MATLAB/Simulink closed-loop control architecture.

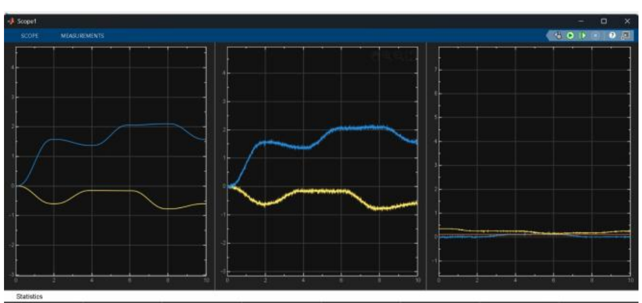


Fig 1.2 — Joints trajectory feedback response.

The computed joint angles are then processed by a Quintic Polynomial Trajectory Planner, which generates smooth joint position, velocity, and acceleration profiles. By ensuring continuity of these motion parameters, the trajectory planner eliminates sudden changes in velocity and acceleration, thereby reducing mechanical stress, minimizing actuator torque peaks, and improving the overall reliability of the robotic system.

The planned trajectory is applied to a Simscape Multibody model, while simulated encoder measurements are processed using an Extended Kalman Filter (EKF). By fusing model predictions with sensor data, the EKF provides accurate joint position and velocity estimates for robust closed-loop control.

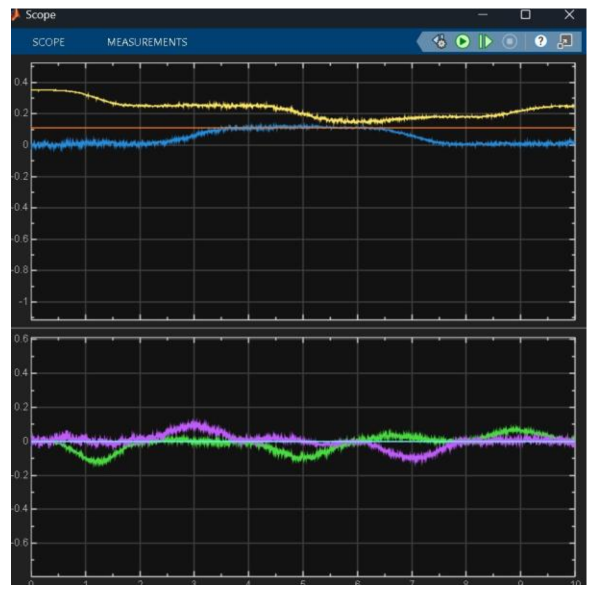


Fig 1.3 — End-effector position and velocity profile.

The estimated joint states are compared with the reference trajectory, and the resulting position and velocity errors are processed by a tuned Proportional-Derivative (PD) controller to generate the desired joint accelerations. These accelerations are supplied to the Inverse Dynamics block, which implements Computed Torque Control (CTC) based on the Euler–Lagrange formulation to compensate for inertia, Coriolis, centrifugal, and gravitational effects, thereby computing the required actuator torques for accurate trajectory tracking. Finally, the estimated joint states are processed through the Forward Kinematics and Jacobian blocks to determine the end-effector position and velocity, enabling real-time monitoring and validation of manipulator performance.

Overall, the developed framework integrates trajectory planning, state estimation, feedback control, dynamic compensation, and kinematic analysis into a unified closed-loop architecture. The modular design closely resembles the control strategy employed in modern industrial robotic manipulators and provides a scalable foundation for future implementation on embedded platforms such as STM32 microcontrollers.

2. POWER FLOW ARCHITECTURE

The SCARA system is powered via a 220VAC mains input, which is first conditioned through a Power Factor Correction (PFC) stage to enhance overall energy efficiency and ensure grid stability. This input is then bifurcated into two distinct DC-DC power domains to minimize electromagnetic interference (EMI). A high-efficiency 1kW LLC Resonant converter provides a robust 48V DC bus dedicated to high-current requirements, specifically driving the primary BLDC motors and the secondary stepper motor. In parallel, a dedicated, highly isolated 12V/120W Flyback converter serves as the isolated logic power supply. This rail is strictly dedicated to powering the main control boards, high-speed communication interfaces, and various sensory inputs, effectively protecting sensitive electronics from motor-switching transients.

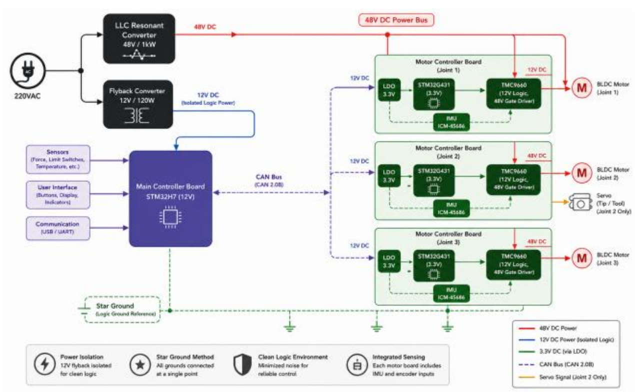


Fig 2.1 — Power flow architecture: 220VAC input to 48V/12V isolated DC domains and joint-level distribution.

To interface with the 3.3V logic levels of the onboard STM32H7 and STM32G4 microcontrollers, local Low-Dropout (LDO) regulators derive a clean 3.3V supply from the 12V rail. This multi-stage isolation—separating the high-voltage/high-current power path for actuators from the low-voltage, noise-sensitive logic path—mitigates risks from back electromotive force (EMF) and high-frequency switching noise, ensuring high integrity and reliability for the entire control architecture.

The system also utilizes a highly integrated TMC9660 motor driver IC. The logic for this IC is powered by the 12V supply, while the MOSFET drive is generated from the 48V rail via the VDRV pin of the TMC9660. The TMC9660 also features low-current outputs at 2.5V, 3.3V, and 5V, with the 5V supply being generated and distributed as required. Additionally, because the FluxCruiser board requires a referenced ground for both the 12V and 48V rails, a star ground is established on the FluxCruiser PCB to reference both the 48V and 12V rails to a common ground.

3. MOTOR DRIVER (FLUXCRUISER)

The FluxCruiser is a distributed motor control PCB deployed at each of the SCARA arm's three kinematic joints, offloading FOC processing to a dedicated gate driver for deterministic, low-latency motion control.

Core Processing & Control Architecture

The STM32G431 microcontroller manages kinematic trajectories, CAN-FD communication, and sensor data, while the Trinamic TMC9660 handles FOC phase mathematics and motor timing, the two linked via a high-speed SPI bus.

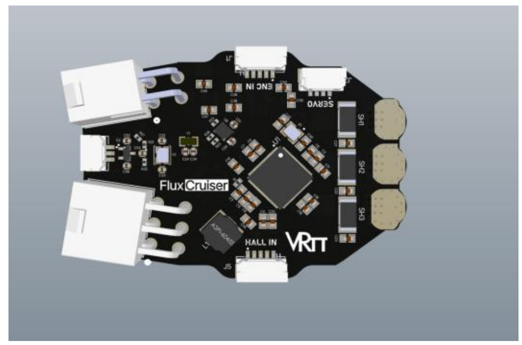


Fig 3.1 — FluxCruiser motor driver board, front view.

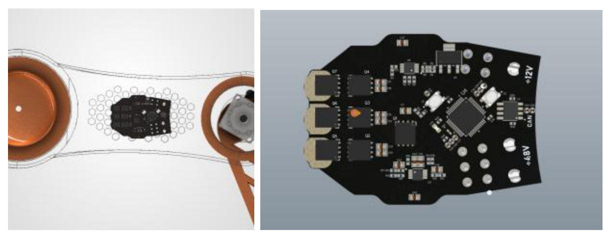


Fig 3.2 — FluxCruiser motor driver board Wireframe and back view.

Power Supply & Distribution

Motor power arrives via a 48V Molex connector rated to 27A, with an isolated 12V flyback rail for logic and gate drivers. An AMS1117-3.3 LDO supplies 3.3V for the STM32G431; the TMC9660's internal LDO generates an auxiliary 5V rail for sensors.

Power Stage, Actuation & Safety

The 3-phase inverter uses over-specced Goodark 100V/80A MOSFETs for thermal headroom against inductive spikes. The STM32G431 also drives an end-effector servo, and a hardware brake chopper dumps regenerative energy into a shunt resistor to protect the 48V bus from overvoltage.

System Communication & Telemetry

The board links to the main STM32H7 via CAN-FD over a JST-GH connector, with JST-GH sensor interfaces for Hall Effect sensors and encoders. An onboard ICM-45686 6-axis

IMU provides real-time spatial data for gravitational compensation and vibration damping.

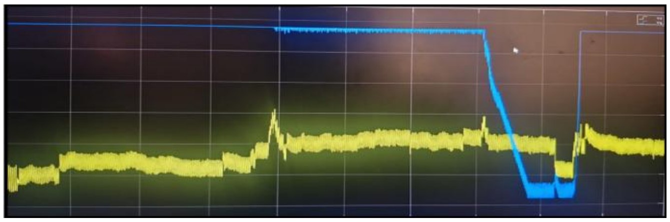


Fig 3.3 — FluxCruiser board, alternate assembly view.

FOC Tuning & Validation

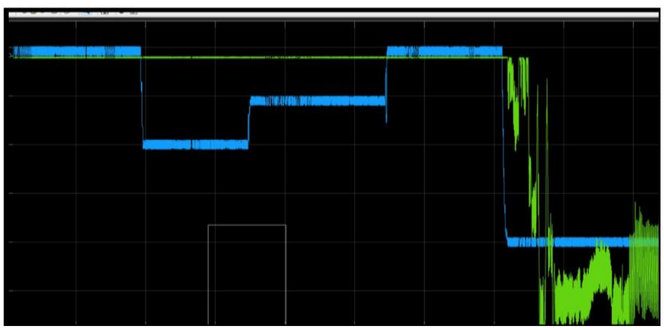


Fig 3.4 — Field Oriented Control PI tuning response.

4. SOFTWARE ARCHITECTURE

A cornerstone of this architecture is its modular software design, which benefits complex robotic systems. By compartmentalizing functions into independent nodes, the system achieves superior maintainability and simplified debugging, as issues are isolated within specific modules. This modularity also allows for seamless component swapping; for instance, a vision algorithm can be updated without altering the motion planning or control stacks. Such flexibility ensures the system remains adaptable to evolving requirements and facilitates collaborative development.

As a proof of concept, we present a Unity3D-based digital twin simulation integrated with ROS2 that replicates SCARA robotic arm behavior. The platform allows developers to test motion, perception, interaction, and control algorithms in a realistic virtual environment before deploying them onto physical hardware.

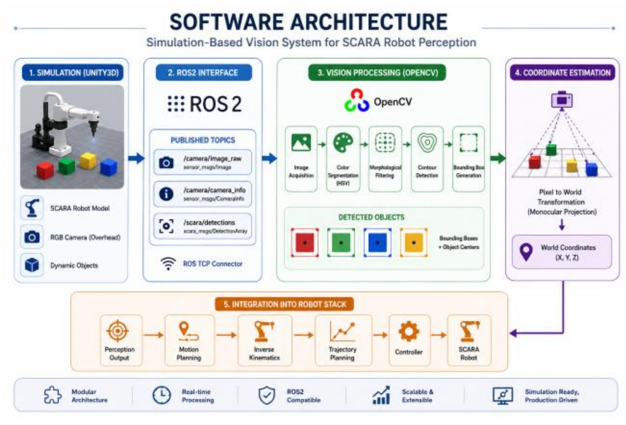


Fig 4.1 — Software architecture: simulation-based vision system for SCARA robot perception.

Simulation Environment

The high-fidelity Unity3D simulation environment serves as a robust digital twin for SCARA robotics, featuring a precise 3D model paired with a fixed overhead RGB camera to effectively mimic real-world industrial inspection systems. The simulation workspace features a customizable ground plane, allowing users to define specific operational boundaries and environmental configurations. To further enhance the fidelity of the digital twin, the environment is configured to mimic real-world unpredictability by introducing simulated sensor noise and dynamic lighting variations. These stochastic elements are crucial for rigorously testing the robustness of perception algorithms against suboptimal conditions like glare and shadows. Validating the vision pipeline in this controlled simulation reduces physical deployment failure risks and ensures predictable performance.

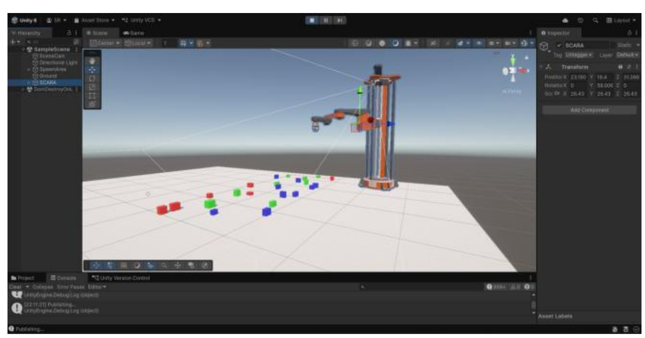


Fig 4.2 — Simulation environment in Unity3D.

ROS2 Integration

The simulation leverages a robust ROS2 integration to facilitate seamless communication between the virtual environment and external processing nodes. Camera data is currently broadcast via the /camera/image_raw topic, providing high-fidelity visual streams for real-time analysis. The system architecture is designed for extensibility, with established pathways for the future integration of advanced topics such as /camera/camera_info and /scara/detections. By decoupling the perception node from the simulation, the framework ensures that image processing operations remain independent and highly performant, allowing for modular development and streamlined testing of robotic control strategies.

Computer Vision & Perception Pipeline

The vision pipeline converts raw camera input into spatial data for the SCARA robot. It begins by publishing RGB images to ROS2, where an OpenCV node performs HSV color segmentation to isolate targets. After applying morphological filtering and contour extraction to refine object boundaries, the system generates bounding boxes to calculate center coordinates. Finally, it uses monocular projection to map these to world (X, Y, Z) coordinates, providing the robot controller with accurate spatial data for real-time pick-and-place operations.

Coordinate Estimation & Mapping

The coordinate estimation module maps visual data to physical space via monocular projection, avoiding costly depth hardware while maintaining accuracy. This data enters the control stack, feeding the motion planner for inverse kinematics. Trajectory planning ensures smooth, collision-free paths for the SCARA manipulator. This integration enables autonomous pick-and-place operations, demonstrating that precise perception, coupled with kinematic planning, provides robust robot control.

Fig 4.3 — Camera feed with bounding box and global coordinates.

This Unity3D-based digital twin architecture mimics production-grade robotic systems, using identical communication protocols to ensure code portability. By decoupling simulation from the control stack, the platform de-risks the transition to physical hardware. This modular, scalable design enables rapid, safe iteration in simulation before committing to real-world deployment.

6. HARDWARE SUMMARY

Supervisory Controller	STM32H743ZIT6 (Vesper Main Controller)
Actuator Node MCU (× 3)	STM32G431
Motor Driver (× 3)	TMC9660 (FOC)
Joint Encoder (× 3)	MSC500 absolute magnetic encoder
Base Offset / Rotation Sensor	PMW3389 optical motion sensor
Height Reference	Magnetic sensor tape
CAN-FD Transceiver	MAX33012EASA+
Ethernet PHY	ADIN1200 (Industrial)
MSC500 Encoder	Absolute magnetic position feedback
PMW3389	Image sensor IC for detecting base offset

Table 51. Core hardware bill of materials for the electronics stack.

6. HARDWARE & SOFTWARE KITS REQUIRED FOR DEVELOPMENT

Simulation & Design Software	Ansys Maxwell — motor and magnetics design, MATLAB/SIMULINK 2025, Altium Designer 2025, SolidWorks 2026
EVM Kits — Main Controller Board	TMC5160-BOB, ADIN1200BCP32Z-R7, CAN_FD MAX33012EASA+
Optical Sensor	PixArt PMW3389 High-Precision Optical Motion Sensor
Magnetic Incremental Encoder	SIKO MSC500 High-Resolution Magnetic Incremental Position Encoder
Motor Driver Board	TMC9660-STP-EVKIT, ADIS16470 IMU kit
Filament	Carbon Fiber Reinforced PLA (CF-PLA)

Table 6.1 — Hardware and software kits required for project prototype

What do you want to achieve on the project if you are selected?

Over the 10-month development window, we want to move from proof-of-concept to a field-tested, self-functional prototype. We'll start by testing and developing the full stack on eval modules of motor control ICs and communication hardware to validate control logic before committing to custom silicon-level bring-up. From there, we'll design, fabricate, and bring up our own PCBs for the modular joint controllers and main controller, and develop our own BLDC actuators around the TMC9660 FOC architecture, tuning the drive stack in-house.

In parallel, we'll experiment with different control algorithms and sensor fusion techniques to build a self-calibrating loop that adjusts actuator input in real time to environmental changes, while testing different SCARA link materials to identify the best strength-to-weight combination for the fastest possible arm movement We would try to implement and calibrate the magnetic sensor system for height mapping and the PMW3389 image-sensor-based offset detection, both of which are crucial for fault detection and prevention.

We'll also focus on power efficiency across varying loads to reduce power draw per hour, and develop an image-detection algorithm to improve positioning precision — enabling the bot to replace manual engraving with a fully autonomous, self-functional process. By the 6-month checkpoint, we aim to have PCB bring-up complete and the base control/self-calibration loop validated on real hardware.

What are your long-term plans to take this project forward?

Beyond the 6-month mark, we want to push the self-calibrating actuator system toward real deployment conditions, complete our in-house BLDC/TMC actuator development, and mature the vision-based precision system to production readiness.

Long term, the team wants to keep iterating on this platform — continuing eval-module-based testing for new components before integration, refining SCARA arm materials for higher speed and lower stress, and potentially extending the system to other precision-automation use cases beyond engraving. Once the core system is proven reliable, we'd like to open it up for wider use or publish our findings. We also plan to develop a UI for bot control and real-time monitoring.

What would be the possible limitations of your proposed solution to the problem?

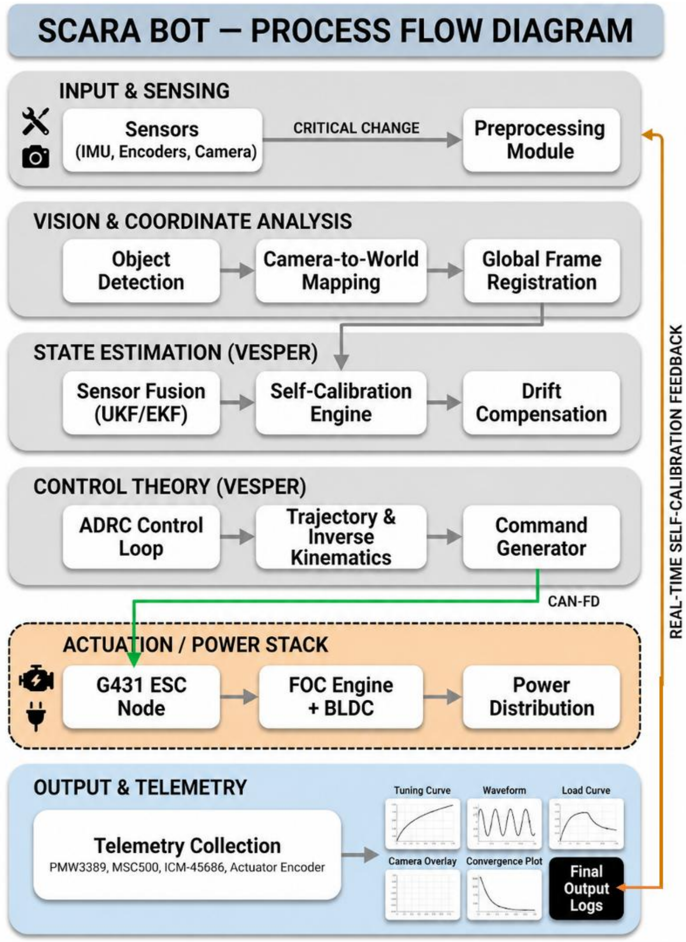
Actuator development is constrained by the need for Ansys (motor/magnetics design) and availability of ADI Trinamic motor ICs, some of which are hard to source in the Indian market, which could slow the actuator timeline even after eval-module validation.

On the sensing side, the onboard IMU shows fluctuations for vibrations and needs to be calibrated. PMW3389 sensor base-offset calibration is non-trivial. The fusion algorithm for the self-error-correction loop — currently planned around a UKF for state estimation with ADRC as the main control algorithm — adds significant tuning and firmware-testing overhead. Mechanically, the middle link currently has less material and is a higher-stress point, so material testing for the fastest possible movement will need to balance speed against structural limits. Direct-drive actuation is still being studied and isn't yet proven for our developed actuators, and camera-based image detection and mapping currently has latency that needs to be reduced for reliable real-time state prediction.

What Hardware, Software, tools, Cloud platforms etc., would you plan to use?

Hardware: ADI-MAX7800X (provided kit), TMC9660-based motor driver boards (starting with TMC9660-STP-EVKIT for eval-stage testing, then custom PCBs), STM32G431/STM32H7 microcontrollers, IMU modules (ADIS16470/ICM-45686-class), and in-house designed PCBs for joint-level and main controllers. We'll also test different structural materials for the SCARA arm links to optimize for the fastest movement without sacrificing rigidity.

Software/tools: MATLAB/Simulink for control-loop design, STM32CubeIDE for firmware, a custom UKF-based sensor fusion implementation, and ADRC for the main control algorithm. Ansys Maxwell will support motor and magnetics design, KiCad/Altium for PCB design, and Unity3D with ROS2 integration plus an OpenCV-based vision pipeline for image detection and precision positioning.



References and Bibliography

The QR-code below consists of a compilation of all the research papers; we have referred too while in development stage of the idea.